

KITEWRIGHT CORE · BUILD PLAN

Kitewright Core prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-04	Amish Chadha	Amish's round-3 decision 7A (KWC-DDR-004); step 11 says how a core moves between frames
0.2	2026-10-03	Amish Chadha	Amish's decision 33B carried out (KWC-DDR-003); carbon plate with countersunk fixings, pocketed rail bars, lighter pin blocks, corner spacers and lid; new strain-relief bar (3.9); power leads now come with the frame (step 11)
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (KWC-DDR-002)

Kitewright Core prototype build plan

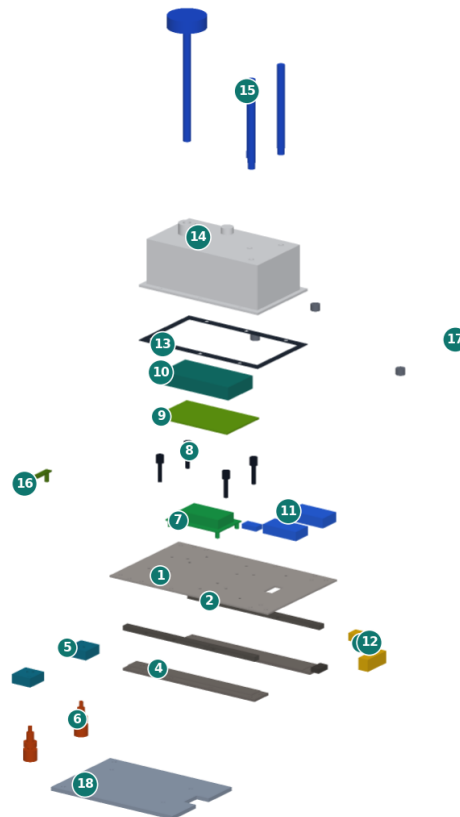
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

■ 1. What you are building

Kitewright Core prototype: every component, pulled apart

Numbered in build order. Seen from the front right and above

- 1 ■ Core plate
- 2 ■ Rail spacer bars (2)
- 3 ■ Front stops (2)
- 4 ■ Rail lips (2) with wear tape
- 5 ■ Pin blocks (2)
- 6 ■ Locking pins (2) and jam nuts
- 7 ■ Power board on 8 mm standoffs
- 8 ■ 25 mm standoffs and dampers (4)
- 9 ■ Damping plate
- 10 ■ Flight controller
- 11 ■ Radio, power modules, receiver
- 12 ■ DS-014 pigtail and plug
- 13 ■ Lid gasket
- 14 ■ Lid with grommets and safety switch
- 15 ■ Antennas, GNSS mast and receiver
- 16 ■ Strain-relief bar on two standoffs
- 17 ■ Corner spacers (4)
- 18 ■ Payload shoe



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github.com/BoujeeEnjinia1701/kitewright-core

Figure 1. Every component of the core, pulled apart and numbered in build order.

The prototype is one Kitewright Core: a 240 x 150 mm carbon fibre plate with the avionics on top under a printed lid, and the payload rail and two locking pins underneath, plus one payload shoe. Figure 1 shows its 18 components in the order you make or fit them. Nine are made: the core plate (cut by a carbon-cutting service from the drawing), and, in a small workshop, the rail spacer bars, front stops, rail lips, pin blocks, payload shoe, lid, damping plate and strain-relief bar. Everything else is bought and fitted: the flight controller, GNSS receiver, power board and modules, radios, antennas, the DS-014 pigtail, the locking pins (indexing plungers), dampers, standoffs and countersunk screws. The power leads and their AS150 plugs are not part of the core: Kitewright Lift and Kitewright Range each

bring their own, and they are soldered in when the core goes into the frame (step 11). The work is sawing, drilling, milling small pockets, tapping and filing aluminium bar, one 3D print, cutting glass-epoxy sheet, and soldering heavy power leads. The parts cost about USD 1,767 for the airborne core from the bill of materials; the ground station kit is bought complete. The core weighs about 0.99 kg.

Safety: The finished core switches lithium pack power of up to 60 V and 200 A. Until section 6 says otherwise, power it only from a current-limited bench supply, never from a pack. Soldering 8 AWG leads needs a 100 W iron or better and gets the wire hot enough to burn: hold it with pliers. Cut aluminium edges and glass-epoxy and carbon dust are hazards: deburr every edge, wear gloves and a dust mask, and cut or file G10 and carbon wet or under extraction. Printing ASA gives off fumes; print in a ventilated space.

■ 2. What changed to make it buildable

The scaffold described the core in words; to build it, each part needed a shape, a material and a fixing. Each change below keeps what the core does and is recorded in decision record KWC-DDR-002.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Payload rail	"A plain rail"	Spacer bar and taped lip on each side, bolted under the plate; 0.3 mm gap above the shoe (Figure 12)	Made from stock bar; the gap comes from stock thicknesses
Locking pin	"A locking pin"	Two spring plungers in blocks under the lips (Figure 14)	Either pin alone holds the payload
End of travel	None	Two front stops between plate and lip (Figure 15)	Lines the pins up with their holes
Payload connector	A connector on the mount	A pigtail through a slot in the plate and a notch in the shoe (Figure 21)	A sliding shoe cannot mate a 40-pin plug reliably
Fixings under the plate	Not shown	M3 countersunk screws set from below, heads flush (Figure 23)	The shoe slides just under the plate; clinch nuts cannot go into carbon
Core plate and rails (decision 33B)	Aluminium plate, solid bars	Carbon fibre plate; windows in the spacer bars and pockets under the lips; thinner pin blocks, corner spacers and lid walls	R9: the core must weigh 1.0 kg or less
Power leads (decision 33B)	Four leads with AS150 plugs in the core	The frame's own harness comes in through the grommets and is tied to a strain-relief bar	R9; the lead lengths differ for each frame anyway
Core to frame	"Bolts into any frame"	Hangs under the deck on four bolts and spacers (Figure 28)	Payload load goes straight into the frame
Flight controller	Not shown	On four dampers and a glass-epoxy plate above the power board (Figure 18)	Vibration isolation
Lid	"Cold-rated enclosure"	Printed lid with flange, gasket and six screws	Clears the stop nuts and the deck opening

■ 3. Making the components

Sizes are in millimetres. Positions are given from the centre of the core plate: "along" means toward the front (the end with the cable slot), "across" means to the left or right of the centre line.

3.1 Core plate

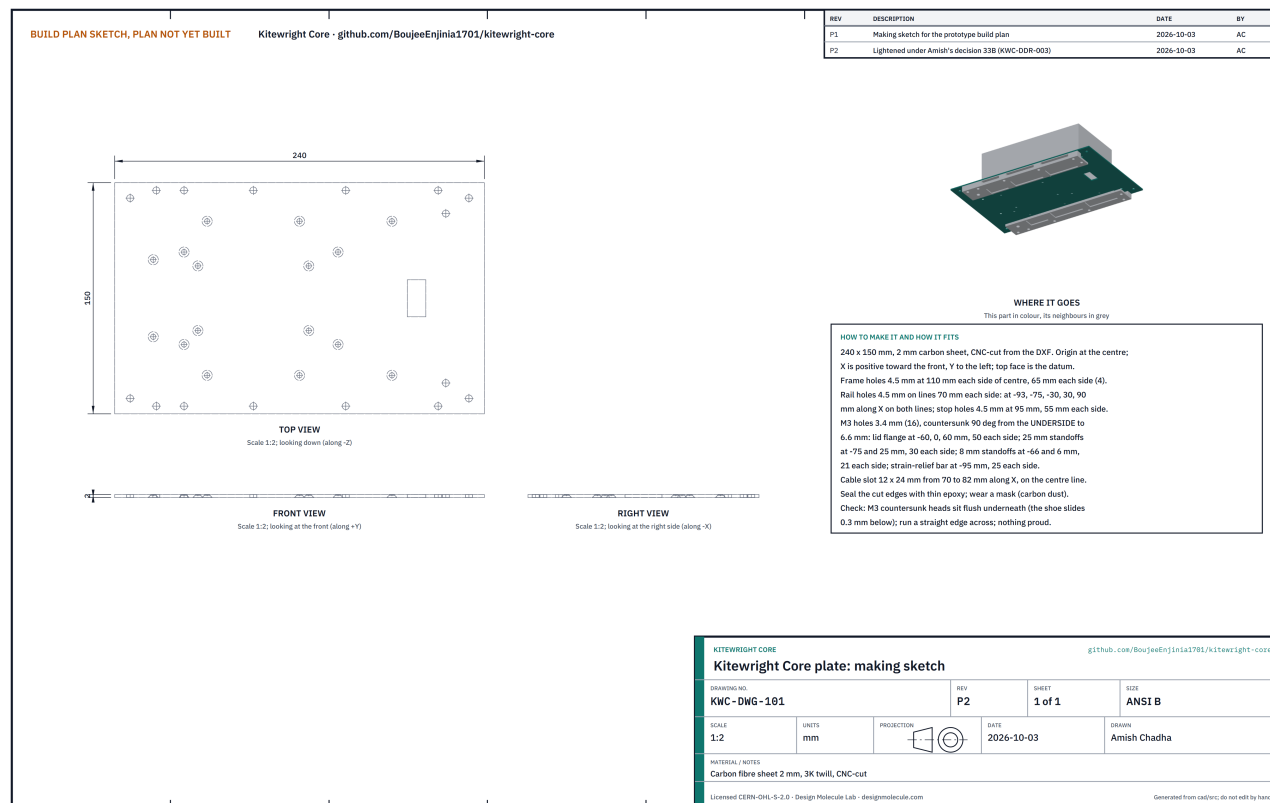


Figure 2. Making sketch KWC-DWG-101, Rev P2.

What it is and what it is made from. The flat plate everything fixes to: 240 x 150 mm of 2 mm carbon fibre sheet (3K twill, quasi-isotropic). Carbon is used because it is light (decision 33B).

How to make it. Order it from a carbon-cutting service (the kind that cuts drone frames), sending the outline and holes below as a drawing file. If you cut it yourself, use carbide tools, cut wet or under extraction and wear a mask.

1. Outline 240 x 150 mm, centre lines marked each way.
2. Four frame holes 4.5 mm, 110 mm along each way from the centre and 65 mm across each way.
3. Rail holes 4.5 mm on two lines 70 mm each side of the centre line, at 93 mm and 75 mm behind the centre, 30 mm behind, 30 mm ahead and 90 mm ahead. Two stop holes 4.5 mm at 95 mm ahead, 55 mm each side.
4. Sixteen M3 holes 3.4 mm, each countersunk 90° from the **underside** to 6.6 mm across so an M3 countersunk head sits flush: six for the lid flange (60 mm behind, on the centre, 60 mm ahead; 50 mm each side), four for the tall standoffs (75 mm behind and 25 mm ahead; 30 mm each side), four for the short standoffs (66 mm behind and 6 mm ahead; 21 mm each side) and two for the strain-relief bar (95 mm behind; 25 mm each side).
5. Cable slot 12 mm long and 24 mm wide, from 70 to 82 mm ahead of the centre.
6. Seal every cut edge and hole with a thin coat of epoxy so the fibres do not fray or take up water.

How it fits the parts next to it. The rails bolt to its underside (Figure 12). The standoffs and the lid's six studs sit on M3 countersunk screws pushed up through it from below with thread-locker (Figures 18 and 23). It hangs from the frame on its four corner holes (Figure 28). Every aluminium part that touches it is anodised, so the two materials cannot corrode each other in wet snow.

Check before moving on. Fit an M3 countersunk screw in each countersink and lay a straight edge across the underside in several directions: nothing stands proud, because the shoe slides 0.3 mm below.

3.2 Rail spacer bars (make 2)

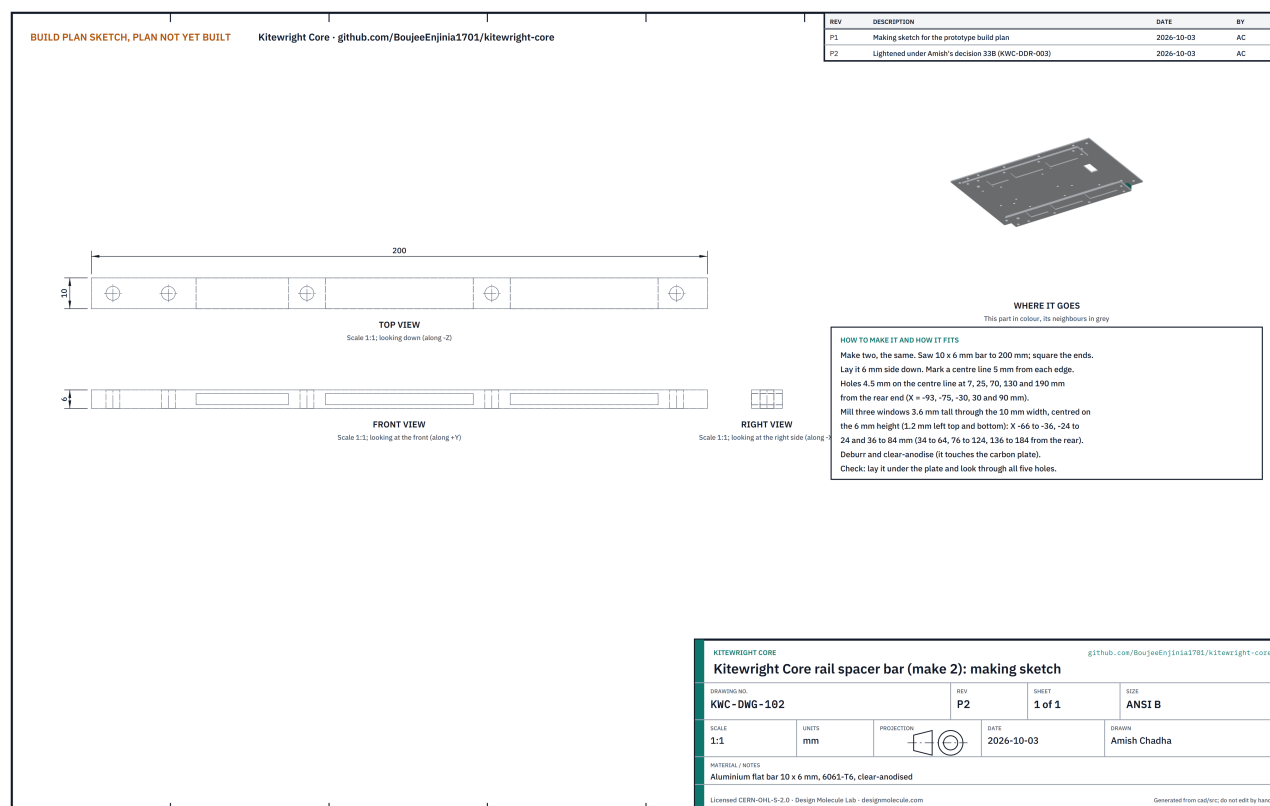


Figure 3. Making sketch KWC-DWG-102, Rev P2.

What it is and what it is made from. The bar that sets the gap between plate and lip: 10 x 6 mm 6061-T6 flat bar, 200 mm long, with three lightening windows.

How to make it.

1. Saw two pieces 200 mm long and square the ends.
2. Mark a centre line along the 10 mm face and drill five 4.5 mm holes on it at 7, 25, 70, 130 and 190 mm from the rear end.
3. Mill three windows right through the 10 mm width, 3.6 mm tall and centred on the 6 mm height, so 1.2 mm of metal is left above and below: 34 to 64 mm, 76 to 124 mm and 136 to 184 mm from the rear end. A small machine shop can do this; it keeps clear of every hole.
4. Deburr, then have both bars clear-anodised (they touch the carbon plate).

How it fits the parts next to it. Its 10 mm face lies against the underside of the plate with its outer edge on the plate's long edge; the lip lies under it (Figure 12). The screws pass through all three.

Check before moving on. Hold it under the plate and look through all five holes.

3.3 Front stops (make 2)

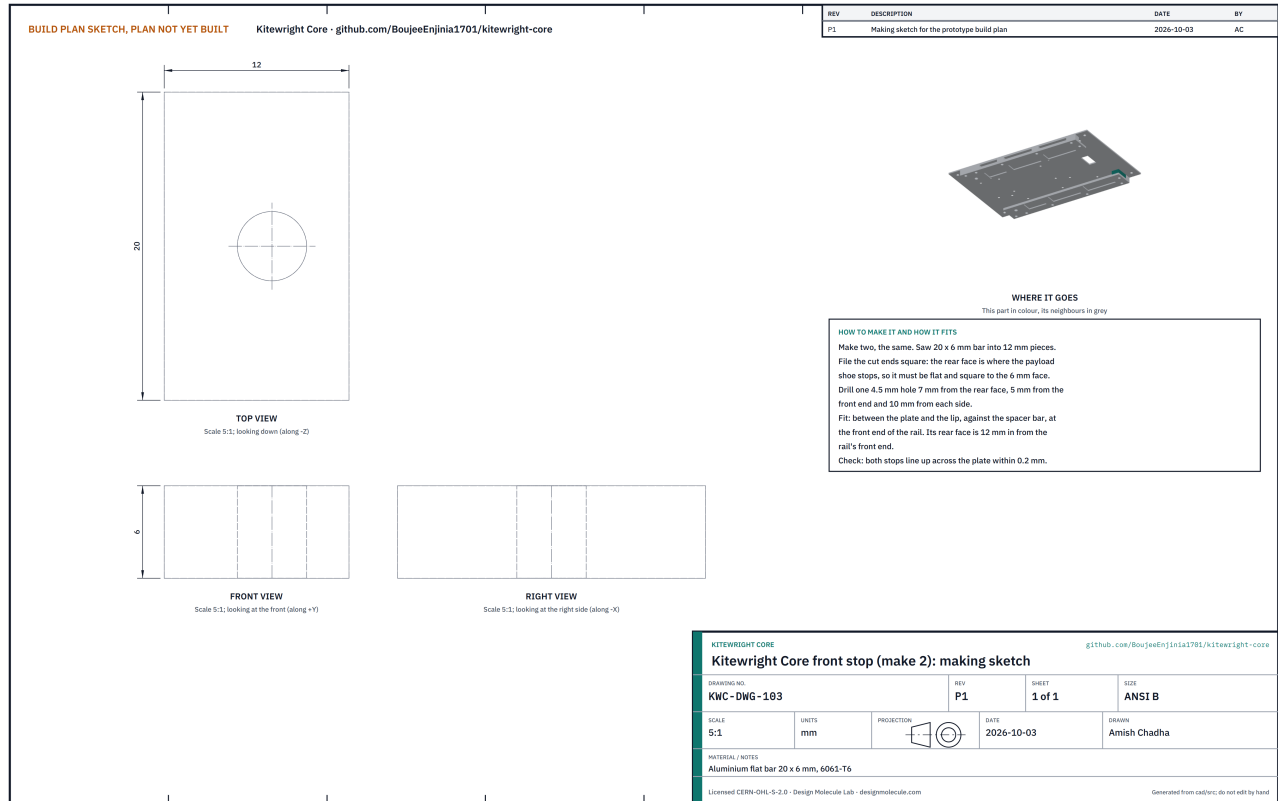


Figure 4. Making sketch KWC-DWG-103.

What it is and what it is made from. A small block that stops the shoe at the front end of the rail: 20 x 6 mm 6061-T6 bar, 12 mm long.

How to make it.

1. Saw two pieces 12 mm long and file the cut ends flat and square: the rear face is where the shoe stops. Have them clear-anodised with the spacer bars.
2. Drill one 4.5 mm hole, 7 mm from the rear face, 5 mm from the front end and 10 mm from each side.

How it fits the parts next to it. It sits between the plate and the lip, against the inner edge of the spacer bar, with its front end level with the rail's front end. The rail's stop screw clamps it (Figure 15).

Check before moving on. With both stops fitted loosely, their rear faces line up across the plate within 0.2 mm.

3.4 Rail lips with wear tape (make a left and a right)

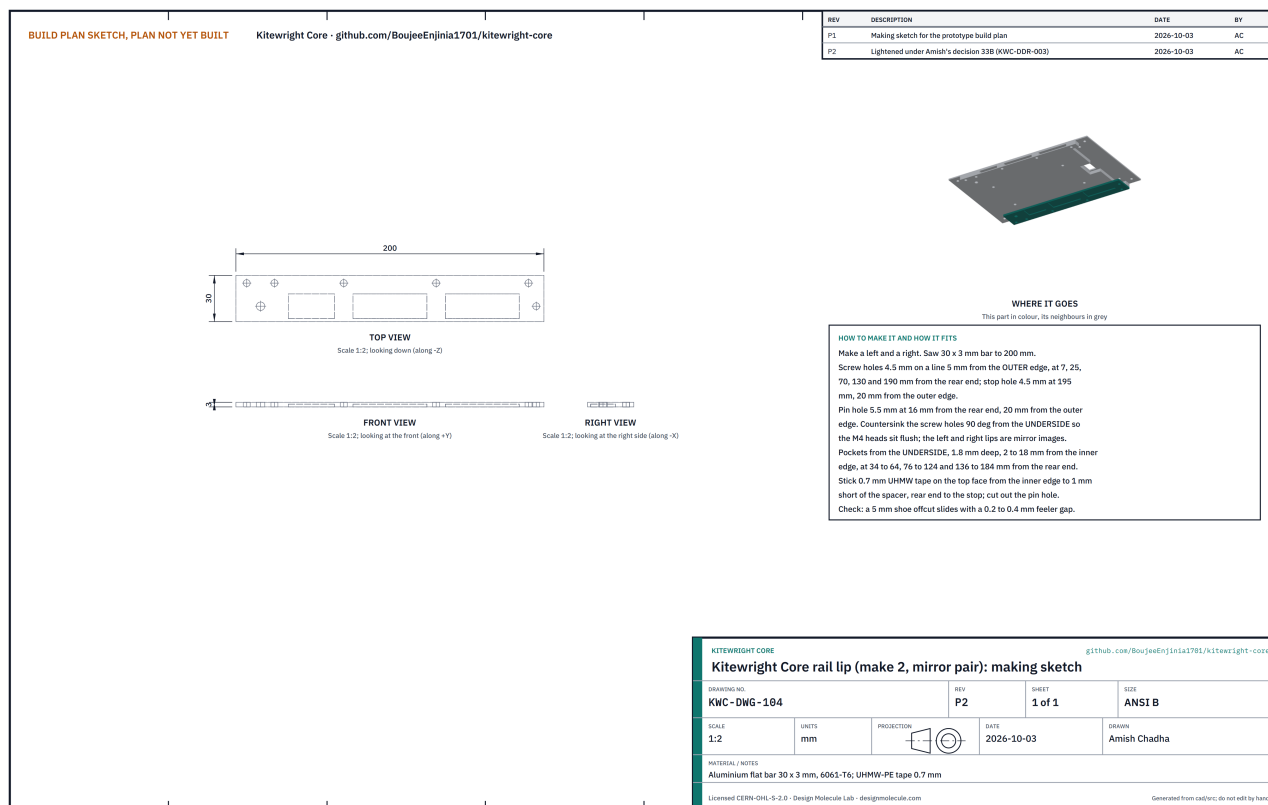


Figure 5. Making sketch KWC-DWG-104, Rev P2.

What it is and what it is made from. The bar the shoe's edge rides on: 30 x 3 mm 6061-T6 flat bar, 200 mm long, with 0.7 mm UHMW-PE tape on top.

How to make it.

1. Saw two pieces 200 mm long.
2. Drill the five screw holes 4.5 mm on a line 5 mm from the outer edge, at 7, 25, 70, 130 and 190 mm from the rear end, and the stop hole 4.5 mm at 195 mm, 20 mm from the outer edge.
3. Drill the pin hole 5.5 mm, 16 mm from the rear end and 20 mm from the outer edge.
4. Countersink the screw and stop holes 90° from the underside, deep enough that an M4 countersunk head sits flush. The left and right lips are mirror images: lay them side by side, outer edges outward, before countersinking.
5. Mill three pockets into the underside, 1.8 mm deep (1.2 mm of metal left), from 2 mm to 18 mm in from the inner edge, at 34 to 64 mm, 76 to 124 mm and 136 to 184 mm from the rear end. They keep clear of the screw line, the pin hole and the pin block.
6. Stick UHMW-PE tape on the top face from the inner edge to 1 mm short of the spacer bar's line, from the rear end to the front stop. Cut out the pin hole with a sharp knife.

How it fits the parts next to it. The tape faces up toward the plate. The shoe's edge overlaps the lip by 19 mm and rides on the tape, with 0.3 mm between the shoe and the plate (Figure 12).

Check before moving on. After step 1 of section 4, a 5 mm offcut of the shoe plate slides between lip and plate along the whole rail, with a 0.2 to 0.4 mm feeler gauge fitting above it.

3.5 Pin blocks (make 2)

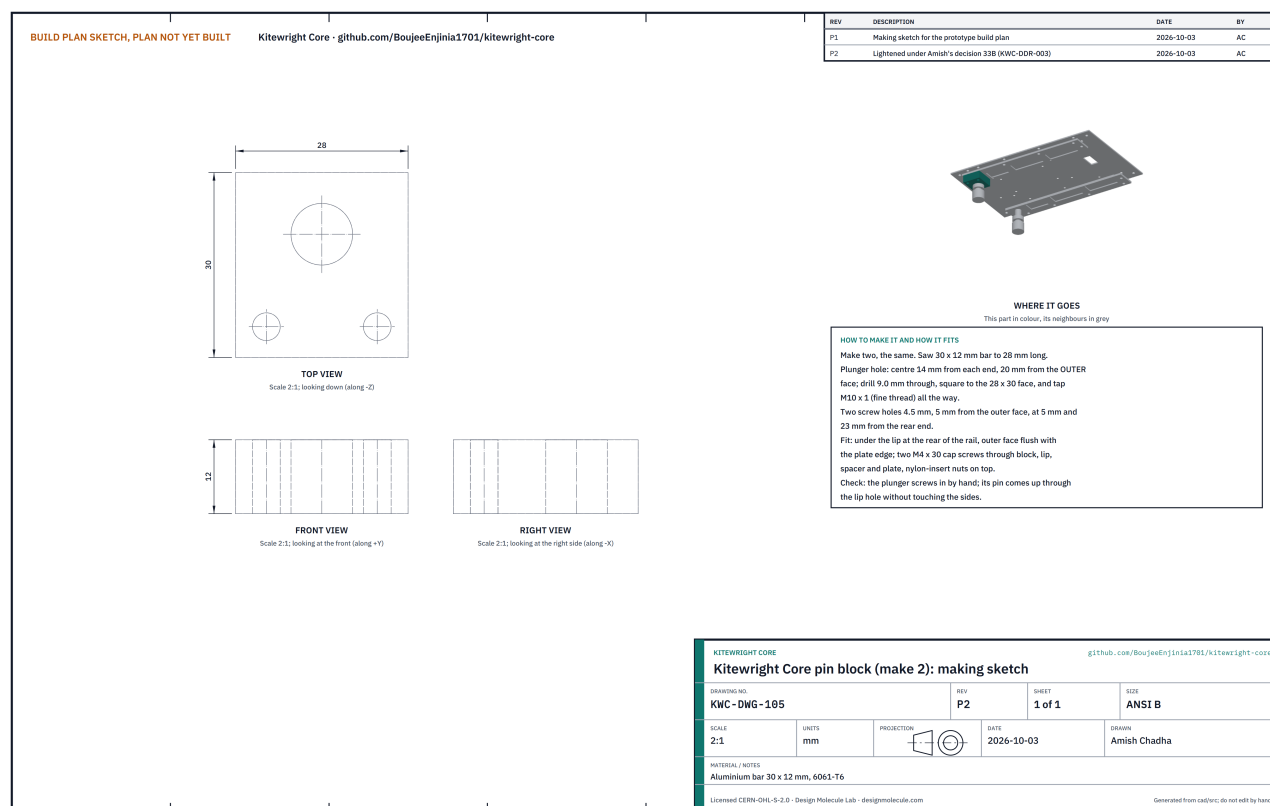


Figure 6. Making sketch KWC-DWG-105, Rev P2.

What it is and what it is made from. The threaded block that holds a locking pin under the lip: 30 x 12 mm 6061-T6 bar, 28 mm long (12 mm of M10 x 1 thread holds the plunger).

How to make it.

1. Saw two pieces 28 mm long.
2. Mark the plunger hole 14 mm from each end and 20 mm from the outer face. Drill it 9.0 mm through, square to the 28 x 30 face, and tap M10 x 1 (fine thread) all the way.
3. Drill two 4.5 mm screw holes 5 mm from the outer face, at 5 and 23 mm from the rear end.

How it fits the parts next to it. The block sits under the lip at the rear of the rail with its outer face flush with the plate's edge; two M4 x 30 cap screws pass up through block, lip, spacer bar and plate into nylon-insert nuts on washers. The plunger screws up into it from below; its pin passes up through the lip and into the shoe (Figure 14).

Check before moving on. The plunger screws in by hand, and its pin comes up through the lip hole without touching the sides.

3.6 Payload shoe

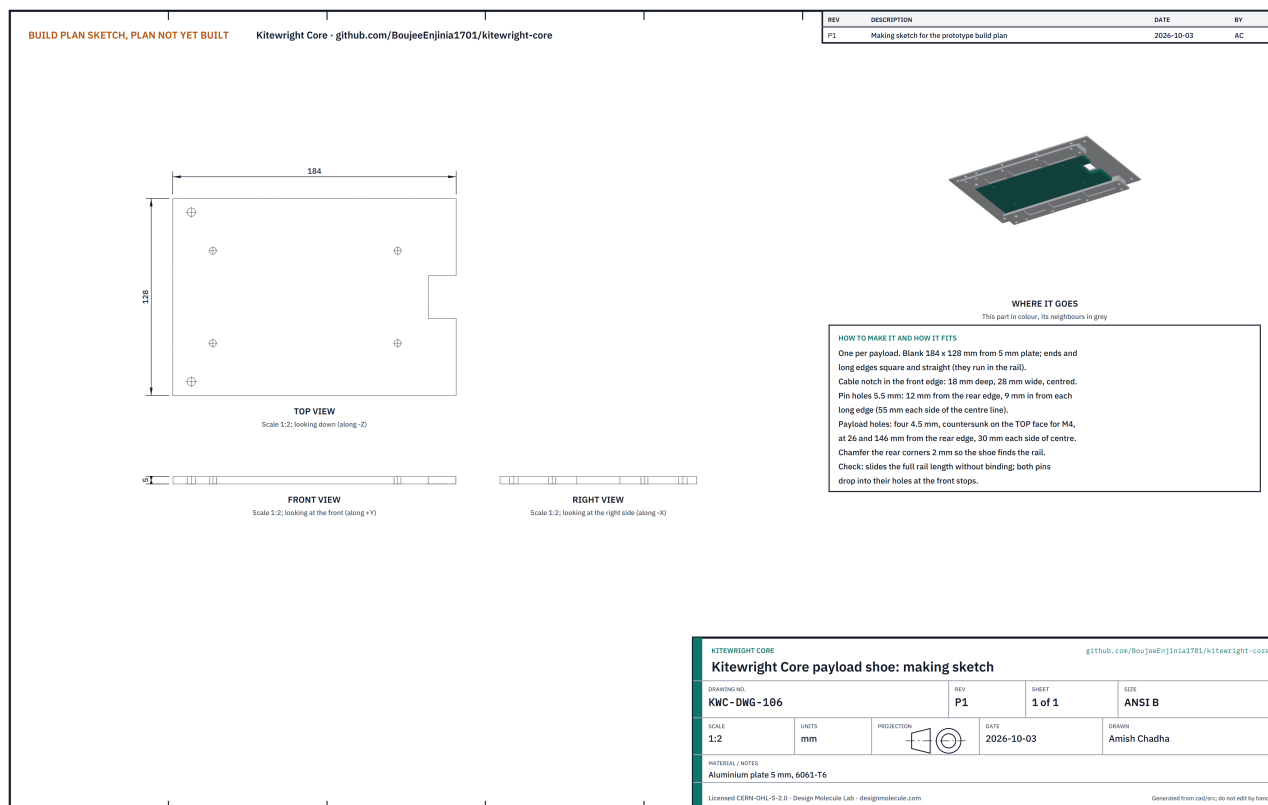


Figure 7. Making sketch KWC-DWG-106.

What it is and what it is made from. The plate every payload carries on its top: 184 x 128 mm, cut from 5 mm 6061-T6 plate. The core kit carries one; each payload maker makes their own to this sketch.

How to make it.

1. Cut the blank 184 x 128 mm. The long edges and the front edge run in the rail and against the stops: make them straight and square.
2. Cut the cable notch in the middle of the front edge, 18 mm deep and 28 mm wide.
3. Drill the two pin holes 5.5 mm, 12 mm from the rear edge and 9 mm in from each long edge.
4. Drill four 4.5 mm payload holes at 26 and 146 mm from the rear edge, 30 mm each side of the centre line, and countersink them from the top so the screw heads sit flush.
5. Chamfer the rear corners 2 mm so the shoe finds the rail.

How it fits the parts next to it. It slides in from the rear between the plate and the lips until its front edge meets the front stops; both pins then rise into its holes (Figures 14 and 15). The payload bolts to its underside with four M4 countersunk screws from the top.

Check before moving on. In section 4, step 10: it slides the full rail without binding, and both pins drop into their holes at the stops.

3.7 Lid

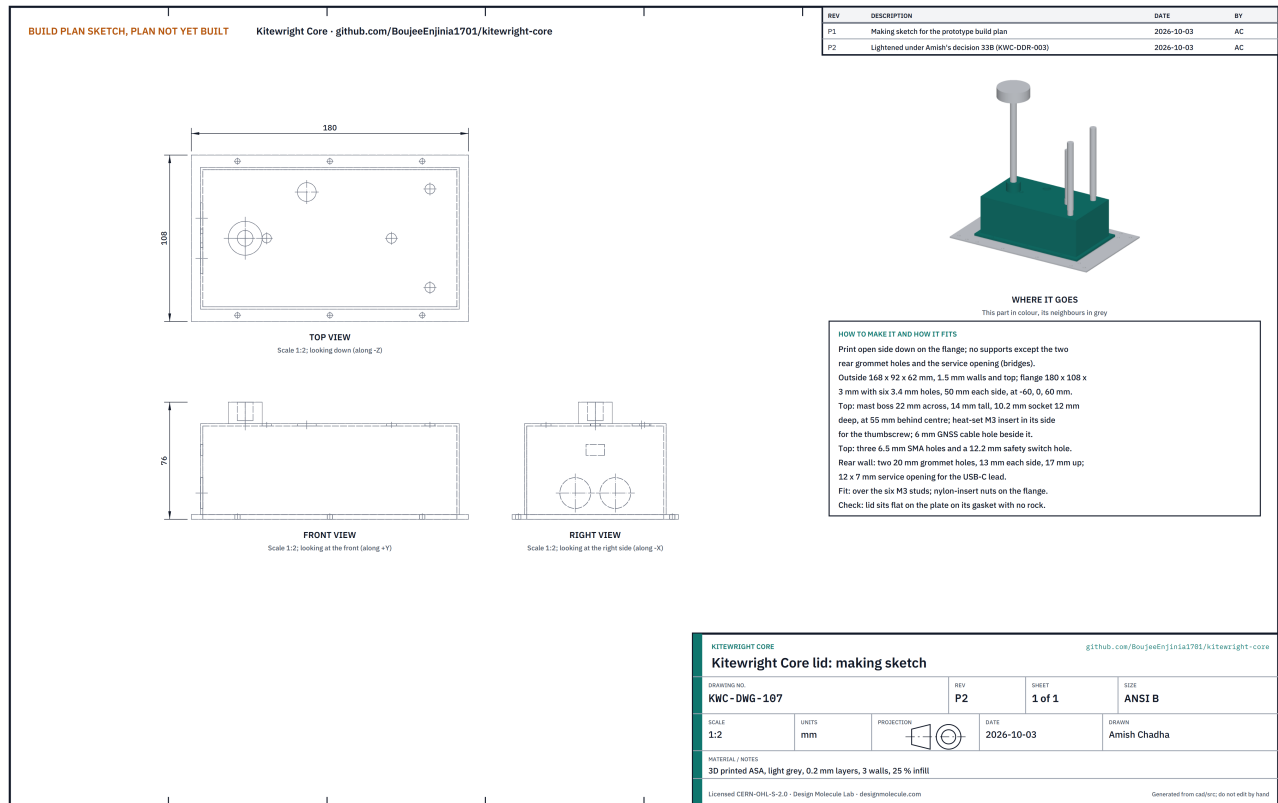


Figure 8. Making sketch KWC-DWG-107, Rev P2.

What it is and what it is made from. The cover over the avionics, printed in light grey ASA: 168 x 92 x 62 mm outside with 1.5 mm walls and top and a 180 x 108 x 3 mm flange.

How to make it.

1. Print it open side down on the flange, 0.2 mm layers, three walls (solid 1.5 mm), 25 % infill, supports only under the rear grommet holes and the service opening.
2. Check the six 3.4 mm flange holes, the 10.2 mm mast socket (12 mm deep in the 22 mm boss, 55 mm behind the centre), the 6 mm GNSS cable hole beside it, the three 6.5 mm antenna holes and the 12.2 mm switch hole in the top, and the two 20 mm grommet holes (13 mm each side, 17 mm up) and the 12 x 7 mm service opening in the rear wall. Clear them with a drill of the same size.
3. Melt an M3 heat-set insert into the side of the mast boss for the thumbscrew.
4. Cut the 1 mm foam gasket to the flange outline, with six 6 mm holes at the screws.

How it fits the parts next to it. The lid stands on the gasket on the plate over six M3 studs (countersunk screws pushed up from below the plate) and is held by six M3 nylon-insert nuts on its flange (Figure 23). It rises through the frame deck's opening, so it can be lifted off from above with the core in the frame.

Check before moving on. The lid sits flat on its gasket with no rock, and the flight controller, once fitted, clears its top by at least 5 mm.

3.8 Damping plate

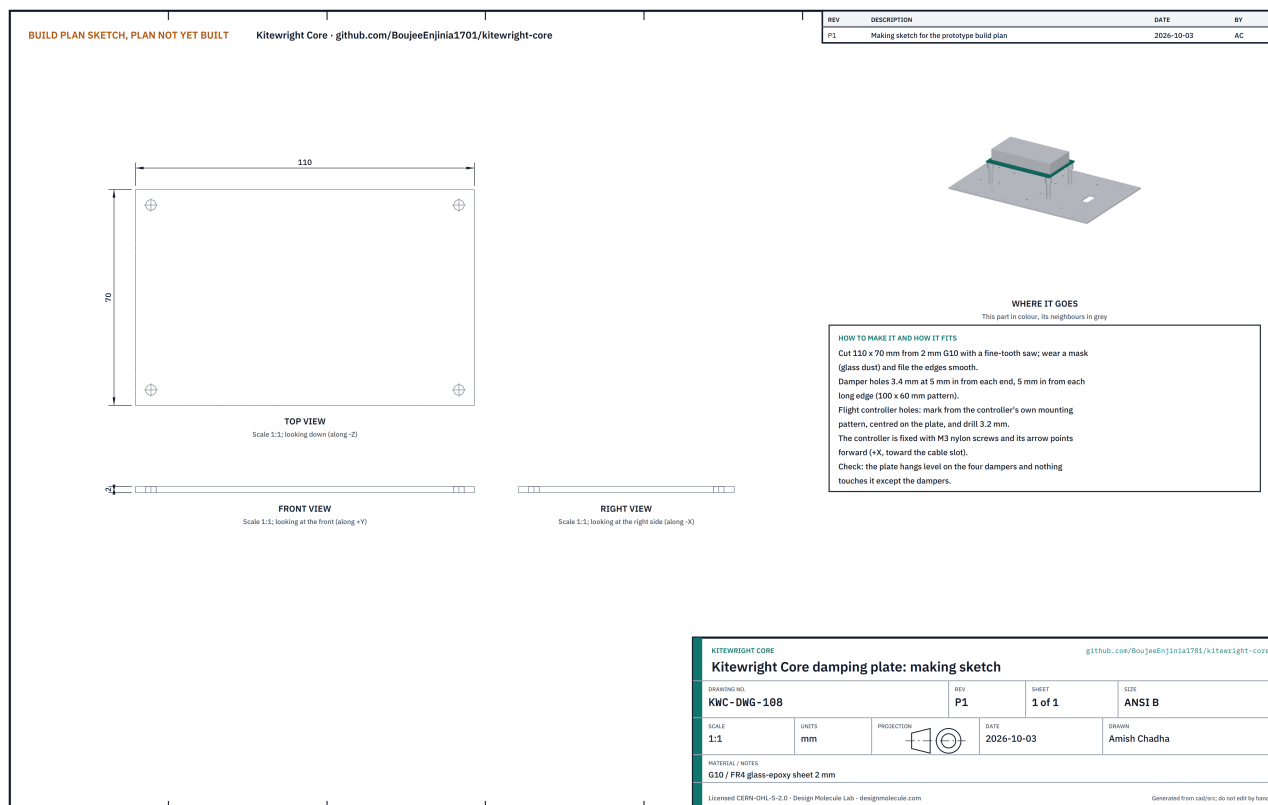


Figure 9. Making sketch KWC-DWG-108.

What it is and what it is made from. The plate the flight controller sits on: 110 x 70 mm of 2 mm G10 glass-epoxy sheet.

How to make it.

1. Cut 110 x 70 mm with a fine-tooth saw, wet or under extraction, and file the edges smooth.
2. Drill four 3.4 mm damper holes 5 mm in from each end and each long edge (a 100 x 60 mm pattern).
3. Mark the flight controller's own mounting pattern, centred, and drill 3.2 mm.

How it fits the parts next to it. It rests on the four silicone dampers, which sit on the tall standoffs; the controller is fixed to it with M3 nylon screws, its arrow pointing forward (Figure 18).

Check before moving on. It hangs level on the dampers and touches nothing else.

3.9 Strain-relief bar

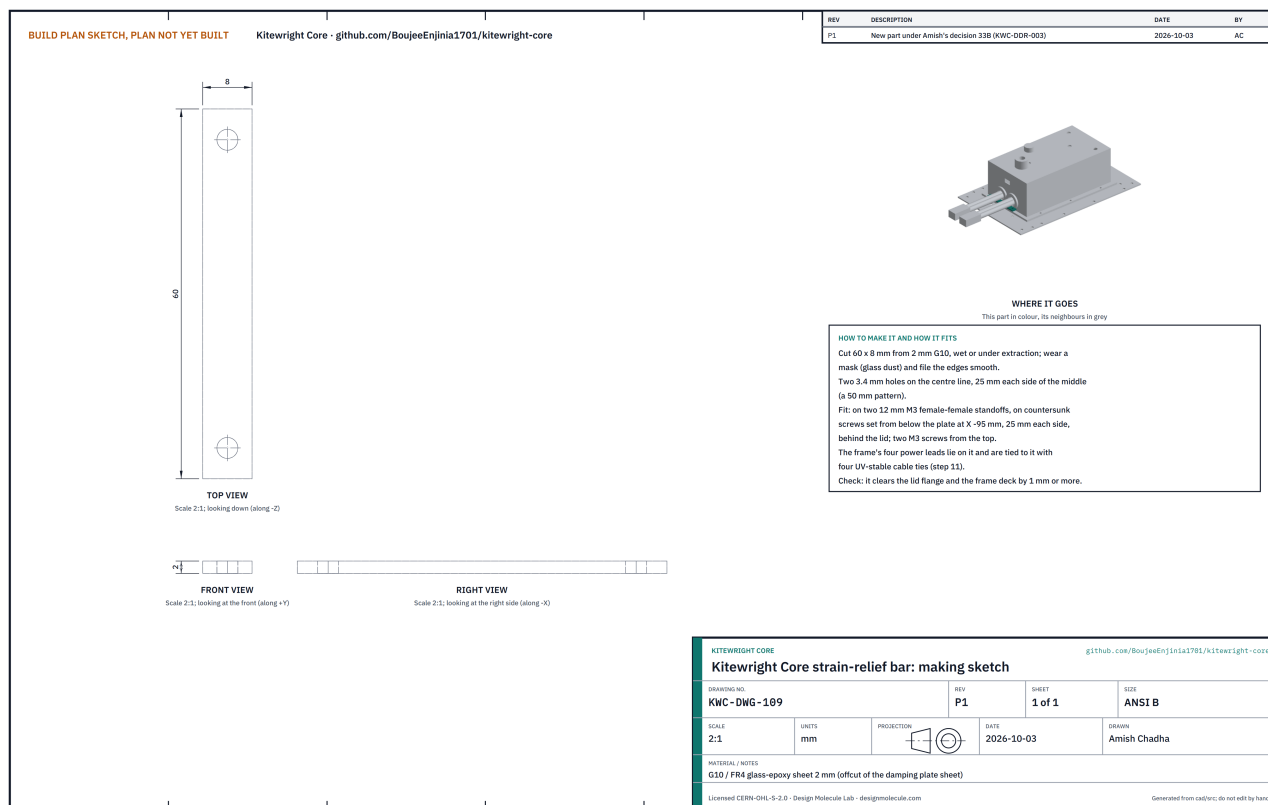


Figure 10. Making sketch KWC-DWG-109 (new under decision 33B).

What it is and what it is made from. A small bar behind the lid that the frame's four power leads are tied to, so a tug on a lead never reaches the solder joints inside: 60 x 8 mm of 2 mm G10, from the damping plate's sheet.

How to make it.

1. Cut 60 x 8 mm, wet or under extraction, and file the edges smooth.
2. Drill two 3.4 mm holes on the centre line, 25 mm each side of the middle.

How it fits the parts next to it. It sits on two 12 mm M3 female-female standoffs, which sit on countersunk screws pushed up from below the plate 95 mm behind the centre and 25 mm each side. The leads lie on it and are tied to it with cable ties (step 11).

Check before moving on. It clears the lid flange and the frame deck by 1 mm or more.

3.10 Bought components

Table 2. Bought components and what to do to them.

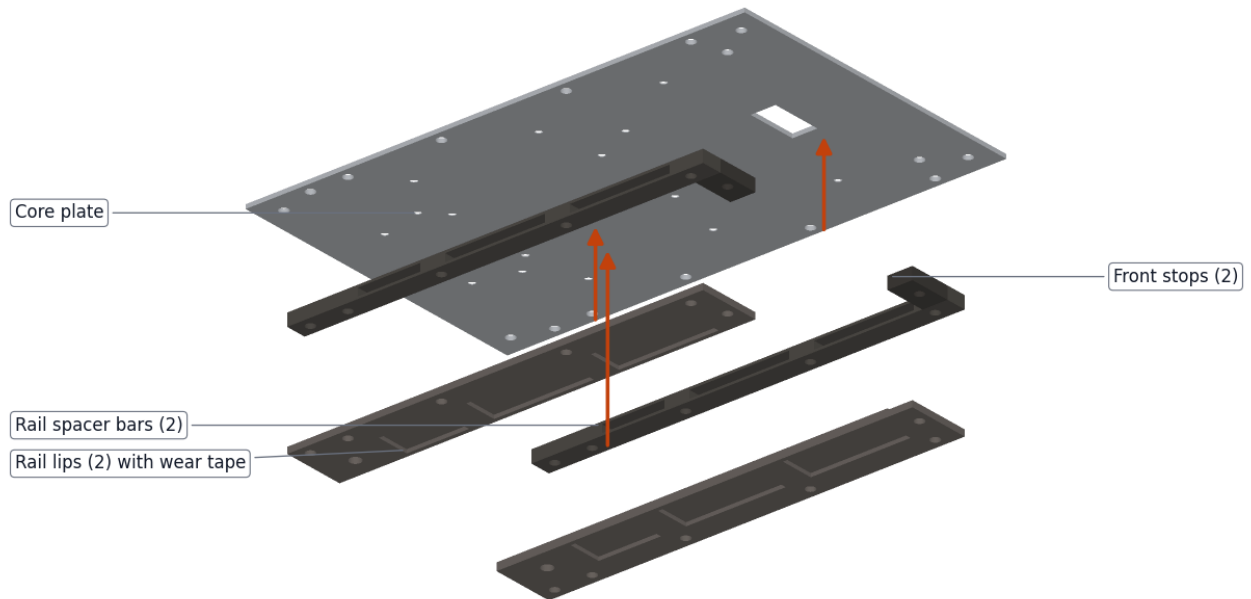
Component	What to buy (specification)	What to do to it
Corner spacers (4)	Aluminium round spacers 12 mm OD x 8 mm, M4 clearance, anodised	Nothing
Locking pins (2)	Stainless indexing plungers, M10 x 1, 5 mm hardened pin, at least 9 mm stroke, pull knob with a rest (lock-out) position, with jam nuts	Paint a red band round each pin just below the lip when the pin is withdrawn, so a withdrawn pin shows red
Countersunk M3 fixings (16)	Stainless M3 countersunk screws, ISO 10642, 8 to 12 mm; six M3 nylon-insert nuts and washers	Push up through the plate's countersinks with thread-locker (steps 3, 4, 7 and 9)
Standoffs (10)	M3 female-female aluminium, anodised: four 25 mm, four 8 mm, two 12 mm (strain-relief bar)	Nothing
Dampers (4)	Silicone grommet dampers for M3, about 10 mm tall, rated -40 °C	Choose hardness for the controller's mass
Flight controller	Pixhawk FMUV6X class with standard baseboard	Load released PX4 and the Kitewright parameter set
GNSS receiver and mast	Multi-band GNSS with compass; 10 mm OD carbon tube, 150 mm; M3 thumbscrew	Fix the receiver to the tube's top with its own mount
Power distribution board	18 to 60 V, 200 A peak, two inputs, two outputs, Hall current sensor, about 80 x 50 mm	Nothing
Power modules	Digital power monitor with 5.3 V supply, backup 5.3 V supply, payload switch (60 V, 8 A)	Nothing
Telemetry radio, receiver, antennas	900 MHz class radio pair, long-range control link receiver, three antennas and SMA bulkhead pigtails	Set the band and power for the country of use
DS-014 pigtail	40-pin connector pair per DS-014, 0.3 m pigtail, cable clamp	Wire to the controller and payload switch per the pin table
Safety switch, grommets, harness, gasket sheet, fasteners	As in <code>bom/bom.csv</code>	Nothing

■ 4. Putting it together

Step 1: rails onto the core plate

Step 1: rails onto the core plate (from below)

Spacer bars, front stops, then taped lips; ten M4 countersunk screws up through all three, nuts on top



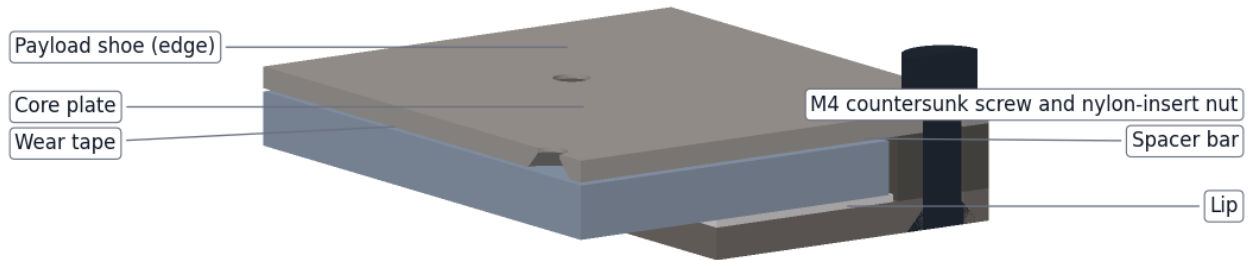
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github.com/BoujeeEnjinia1701/kitewright-core

Turn the plate upside down. Lay the two spacer bars along its long edges, the front stops at the front against their inner edges, then the taped lips, outer edges flush. Push ten M4 x 16 countersunk screws up through lip, spacer bar and plate (and the two stop screws through lip, stop and plate), and fit nylon-insert nuts on washers on top. Use thread-locker; tighten to about 2 N·m, no more, so the washers do not crush the carbon. **Hold point:** do the shoe-offcut and feeler check of 3.4 now.

Joint 1: the plain rail, cut across at a screw

Shoe edge runs between plate and taped lip: 0.3 mm gap above, 1 mm at the side



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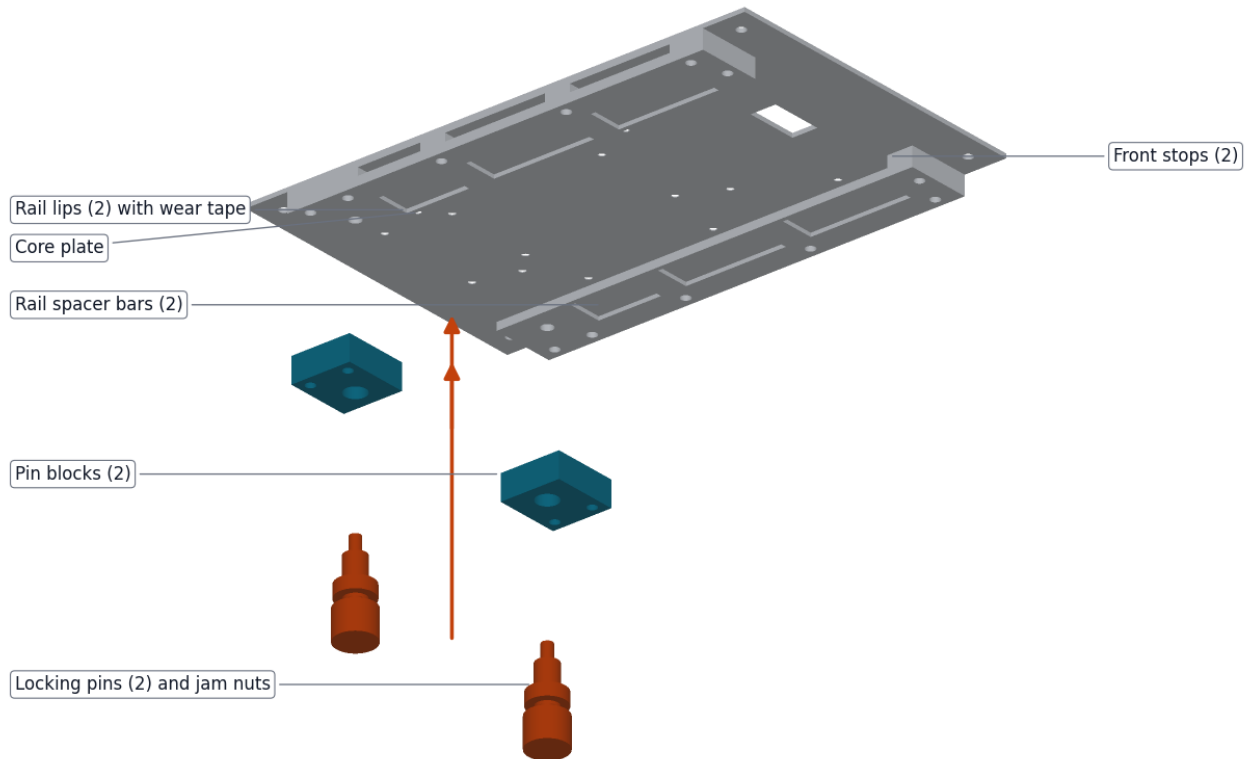
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Figure 12. The rail cut across at a screw: the shoe's edge runs on the taped lip, 0.3 mm under the plate.

Step 2: pin blocks and locking pins

Step 2: pin blocks and locking pins

Blocks under the lips at the rear, M4 x 30 screws; screw each plunger in until its pin is 0.5 mm short of the plate



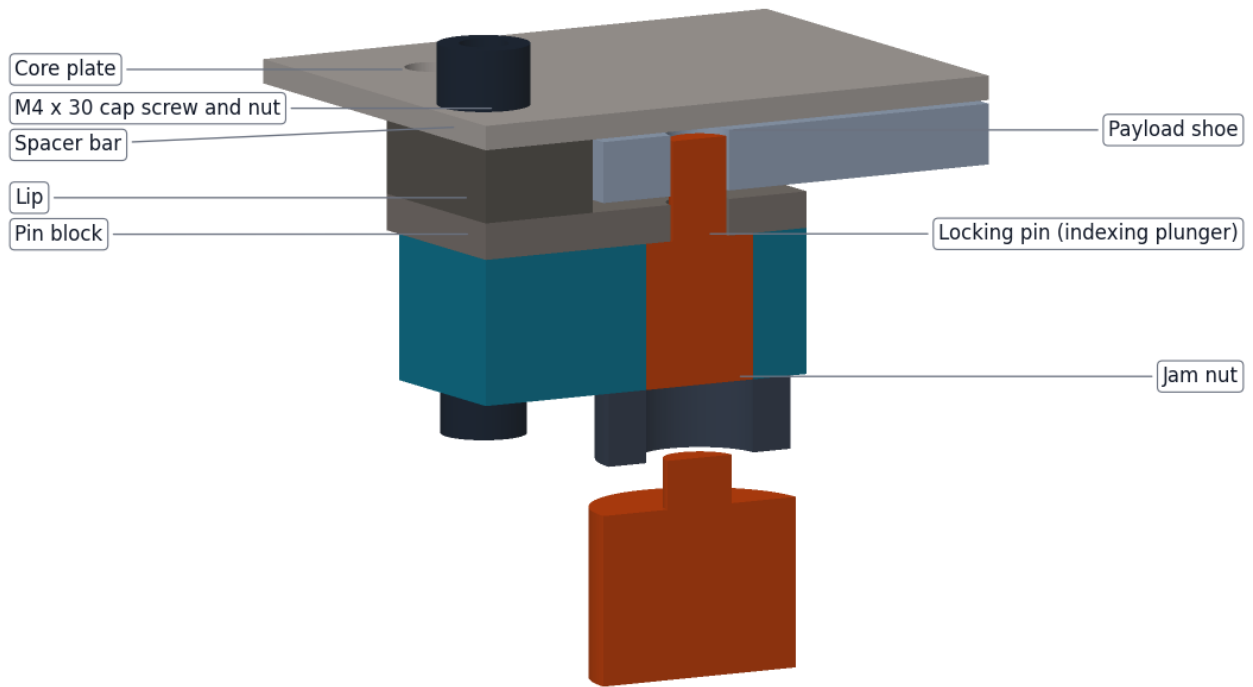
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Fit each pin block under its lip at the rear with two M4 x 30 cap screws up through block, lip, spacer bar and plate, nuts on top. Screw a plunger up into each block until, released, its pin tip is 0.5 mm short of the plate; lock it with the jam nut.

Joint 2: a locking pin, cut through its axis

Pin passes up through the lip into the shoe; pull the knob down and twist to hold it out



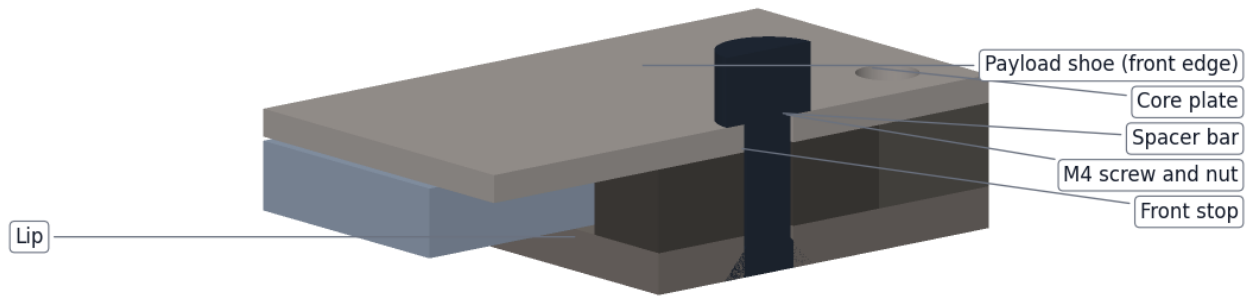
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Figure 14. A locking pin cut through its axis: the pin rises through the lip into the shoe.

Joint 3: a front stop, cut through its screw

The shoe's front edge stops against it; both pins then line up with their holes



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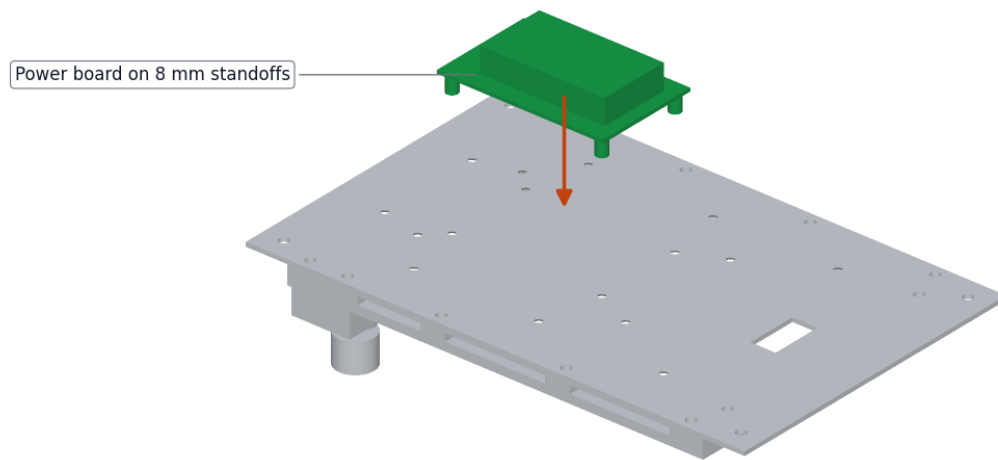
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Figure 15. A front stop, clamped between plate and lip by the stop screw.

Step 3: power board

Step 3: power board

8 mm standoffs on countersunk screws set from below; board on top; tin the four pads for the frame's leads



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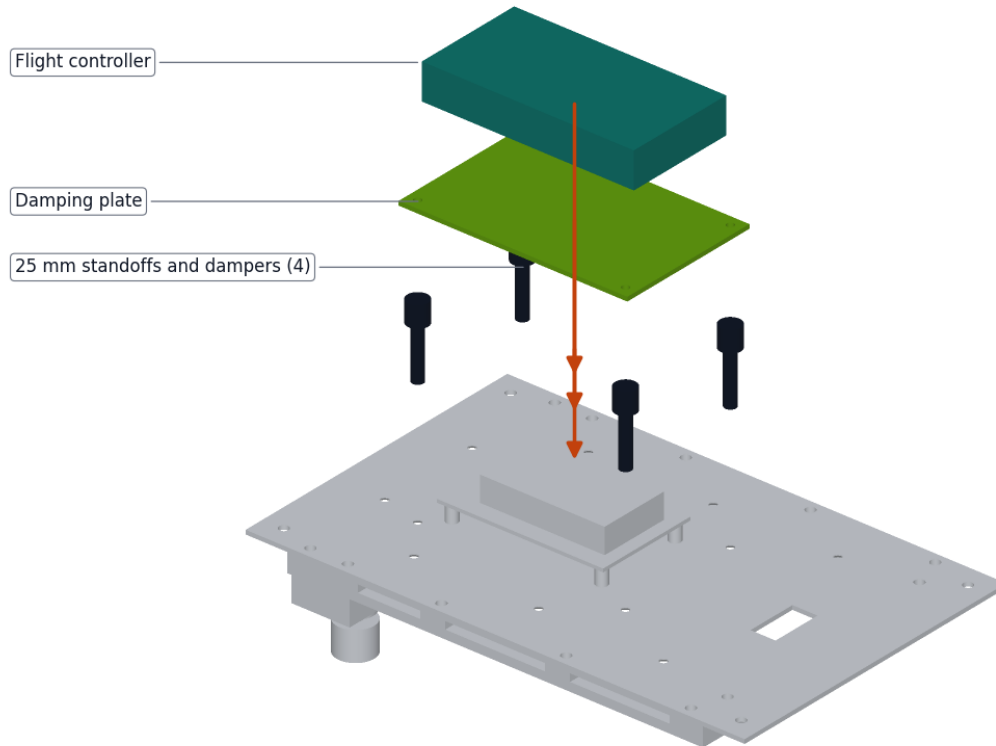
github.com/BoujeeEnjinia1701/kitewright-core

Turn the plate the right way up. Push four M3 countersunk screws up through the short-standoff countersinks with thread-locker, screw the four 8 mm standoffs onto them and fix the power board on top with M3 screws. Tin the board's four big pads (two pack inputs, two frame outputs); the frame's own leads are soldered to them in step 11.

Step 4: flight controller on its dampers

Step 4: flight controller on its dampers

25 mm standoffs, dampers, damping plate; controller on M3 nylon screws, arrow forward



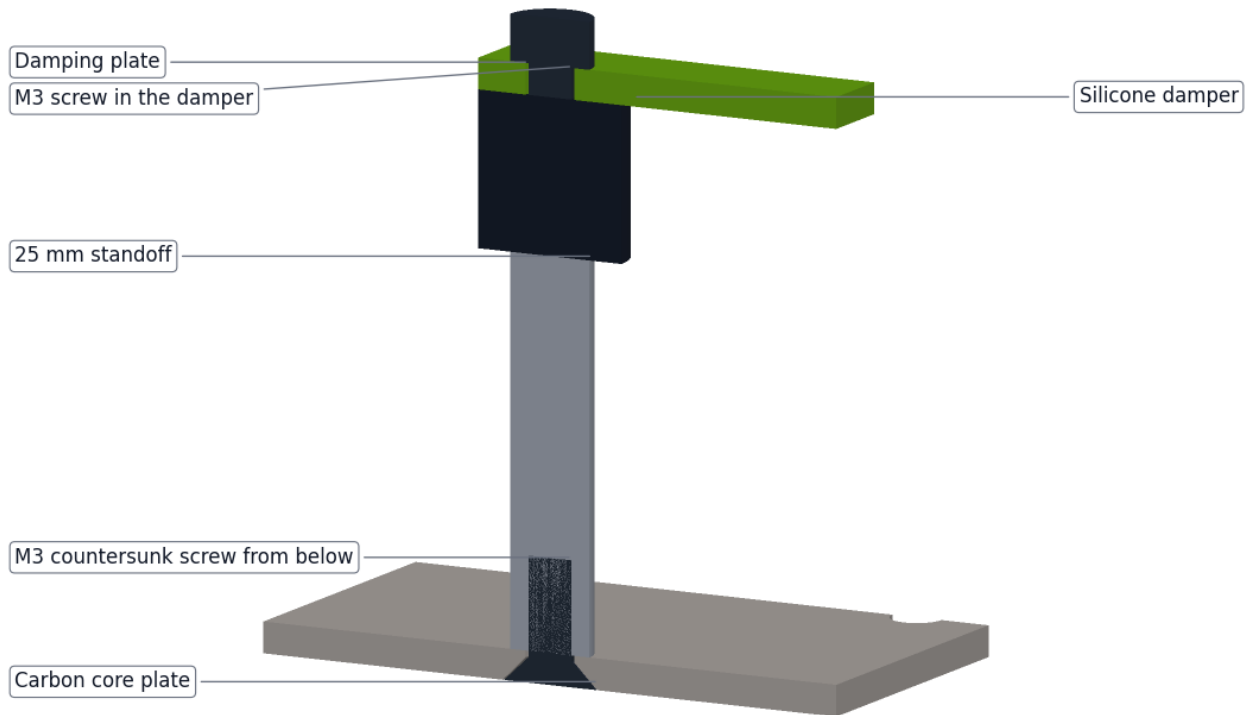
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Push four M3 countersunk screws up through the tall-standoff countersinks with thread-locker and screw the four 25 mm standoffs onto them. Fit a damper on each, lay the damping plate on the dampers and fix it with M3 screws into the dampers. Fix the flight controller to the plate with M3 nylon screws, arrow forward.

Joint 5: flight controller damping, cut through a damper

Standoff on a flush countersunk screw set from below; the plate floats on four silicone dampers



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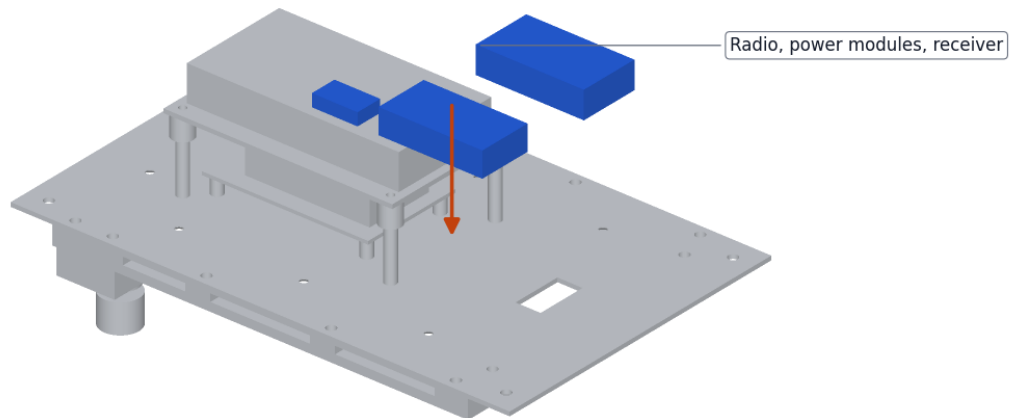
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Figure 18. Flight controller damping, cut through a damper.

Step 5: radio, power modules and receiver

Step 5: radio, power modules and receiver

Foam-taped to the plate in front of the controller; plug the harness



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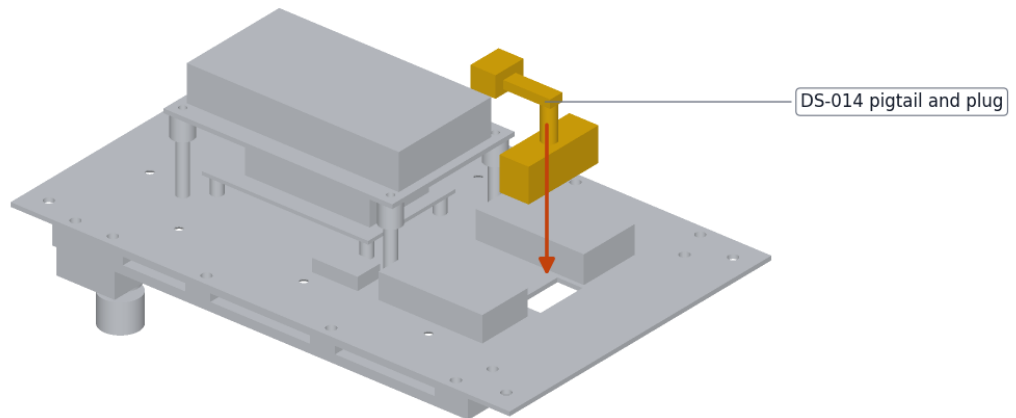
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Stick the telemetry radio (left front), power modules (right front) and control link receiver (right, under the damping plate) to the plate with foam tape, at room temperature. Plug in the harness: power modules to the controller's two power inputs, radio to a telemetry port, receiver to the control input, power board current sensor to its input, payload switch enable to a spare output.

Step 6: DS-014 pigtail

Step 6: DS-014 pigtail

Cable down through the slot, clamped to the plate; plug left hanging below



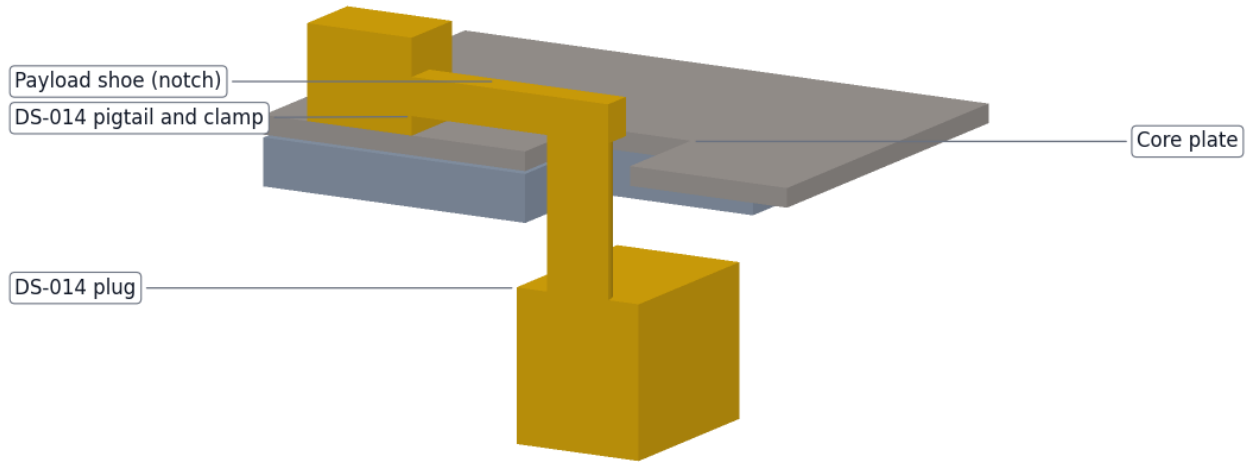
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-core

Pass the pigtail's plug down through the cable slot from above and clamp the cable to the plate just behind the slot. Wire its power pins to the payload switch and its data pins to the controller's Ethernet, CAN, serial and trigger ports, per the DS-014 pin table.

Joint 4: the DS-014 pigtail, cut on the centre line

Cable drops through the 12 x 24 mm slot and the shoe's notch; plug hangs in front of the payload



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

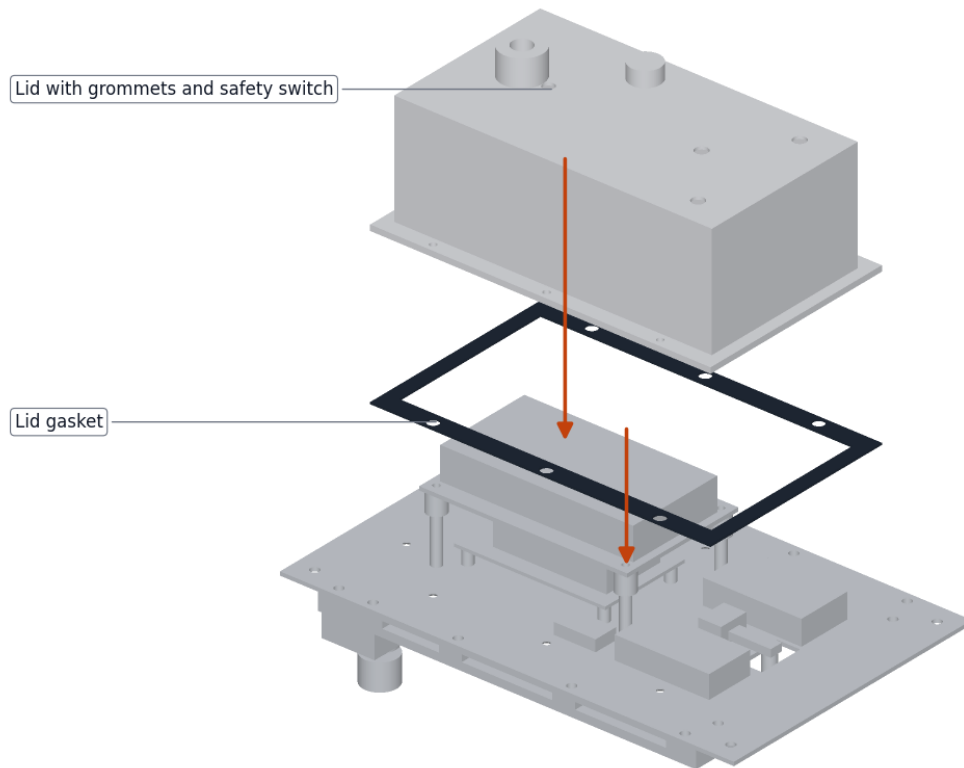
github.com/BoujeeEnjinia1701/kitewright-core

Figure 21. The pigtail through the slot and the shoe's notch; the plug hangs in front of the payload.

Step 7: gasket and lid

Step 7: gasket and lid

Grommets in the rear wall; lower the lid over the six studs; six M3 nylon-insert nuts



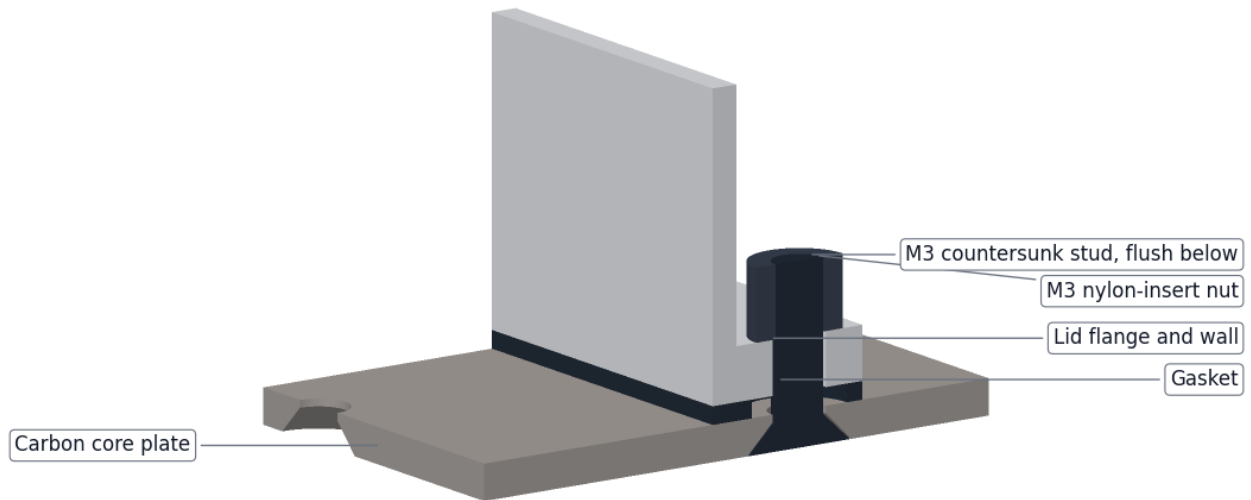
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-core

Push six M3 x 12 countersunk screws up through the flange countersinks with thread-locker: they are the lid's studs. Stick the gasket to the plate around the avionics. Fit the two grommets to the lid's rear wall. Lower the lid over the studs onto the gasket and fit six M3 nylon-insert nuts on washers, snug but not crushing the gasket.

Joint 6: lid flange to plate, cut through a screw

Six countersunk studs from below, nuts on the flange; the 1 mm foam gasket seals it



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

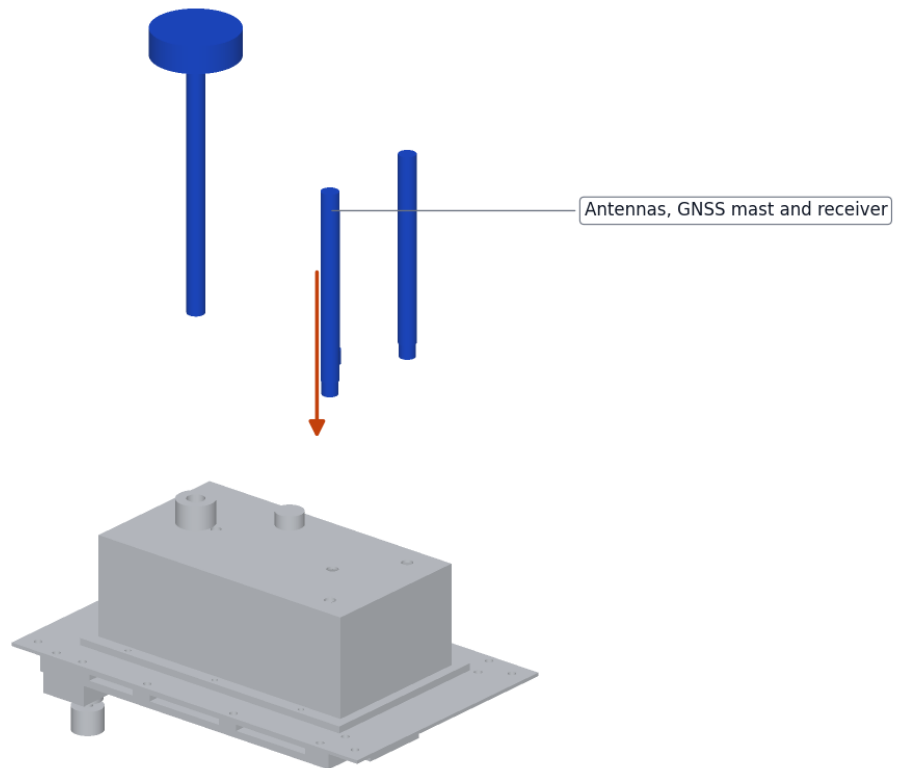
github.com/BoujeeEnjinia1701/kitewright-core

Figure 23. Lid flange to plate: the flush countersunk stud, gasket and nut.

Step 8: antennas, GNSS mast and receiver

Step 8: antennas, GNSS mast and receiver

SMA bulkheads through the lid top; mast into its socket, thumbscrew



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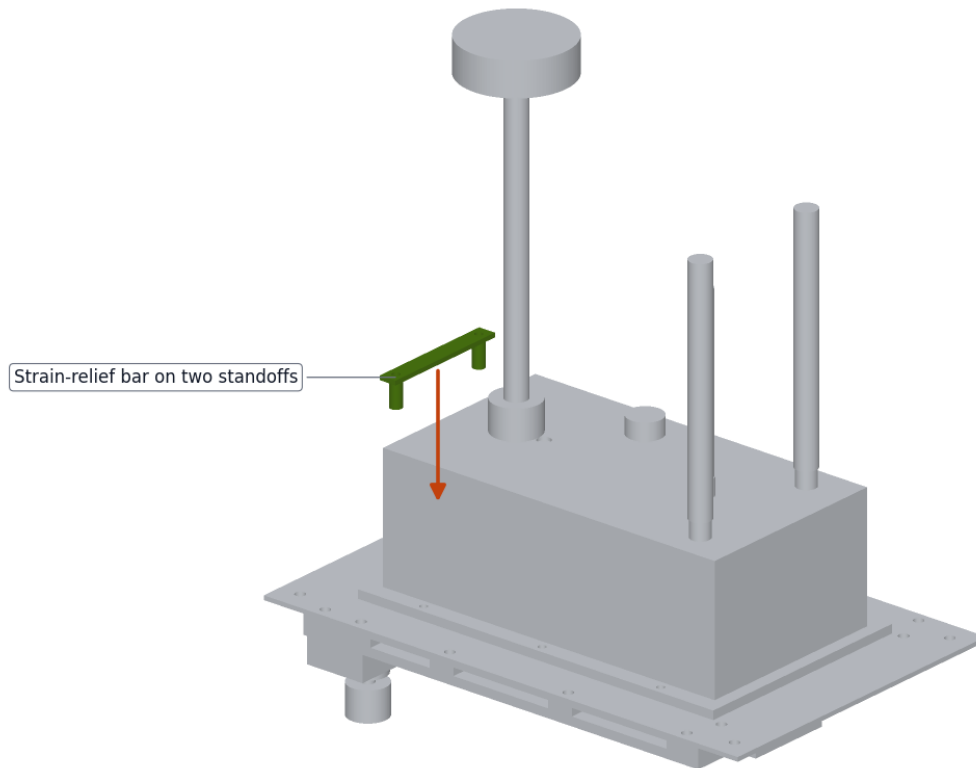
github.com/BoujeeEnjinia1701/kitewright-core

Fit the three SMA bulkheads through the lid top and screw on the antennas. Press the safety switch into its hole. Push the mast into its socket, lock it with the thumbscrew, and run the GNSS cable in through the hole beside the boss.

Step 9: strain-relief bar

Step 9: strain-relief bar behind the lid

Two 12 mm standoffs on countersunk screws set from below; G10 bar on top with two M3 screws



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

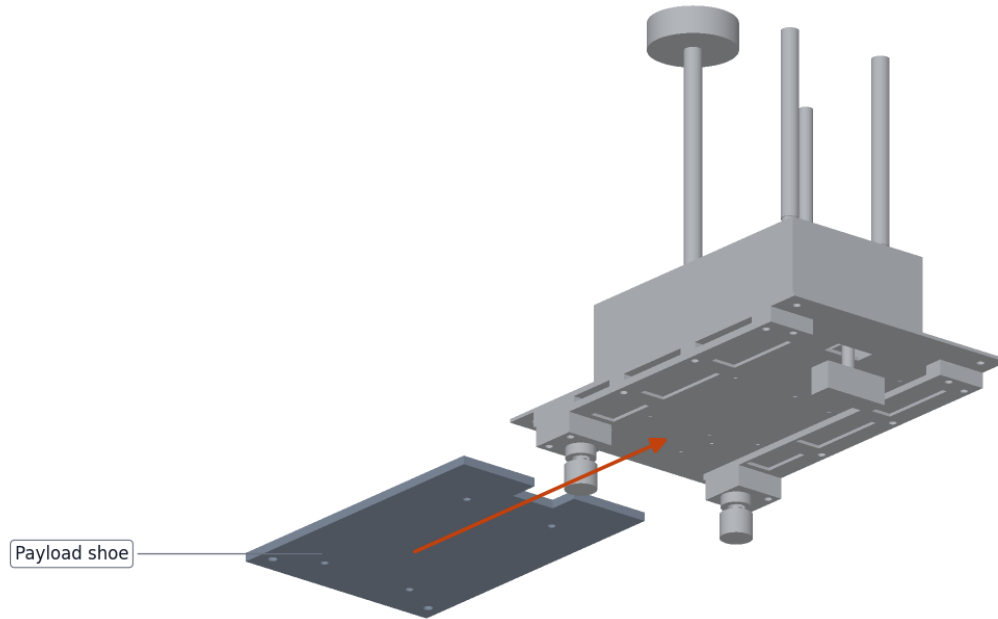
github.com/BoujeeEnjinia1701/kitewright-core

Push two M3 countersunk screws up through the countersinks behind the lid with thread-locker, screw the two 12 mm standoffs onto them, and fix the G10 bar on top with two M3 screws. The cable ties go on in step 11.

Step 10: payload shoe in from the rear (payload swap)

Step 10: payload shoe in from the rear (payload swap)

Pull and twist both pins out, slide to the front stops, twist back: both pins snap in, no red showing



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-core

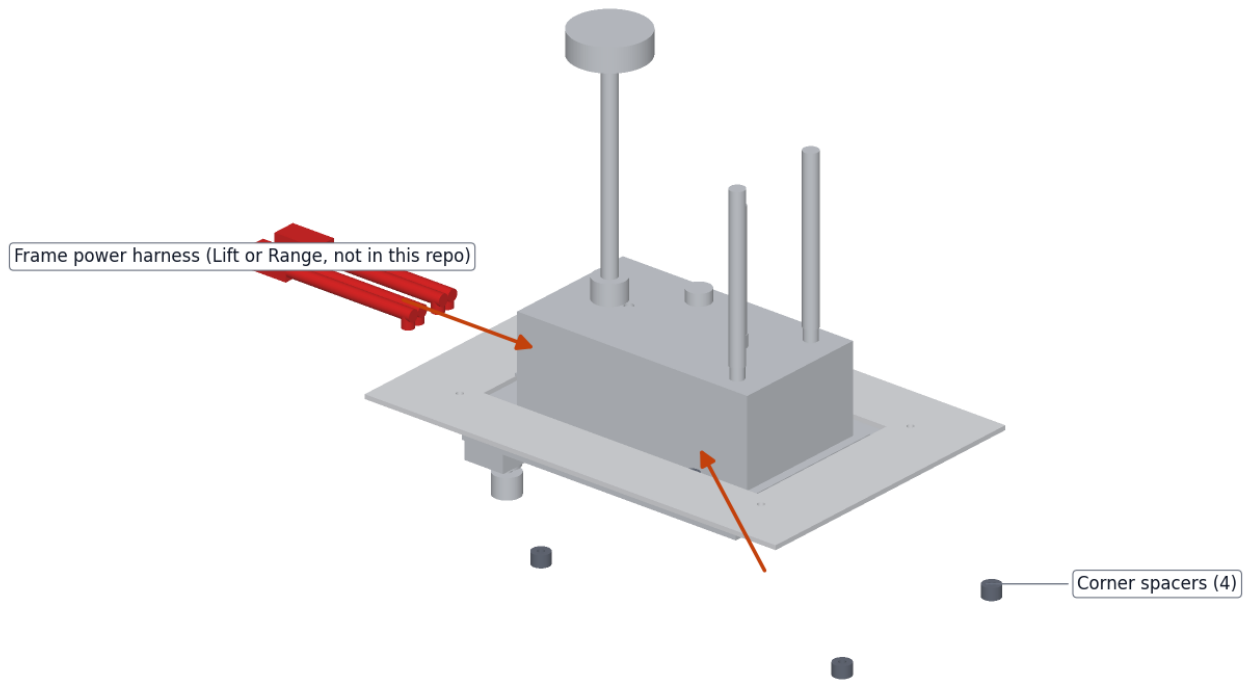
Pull both plunger knobs down and twist them to their rest. Slide the shoe into the rail from the rear until it meets the front stops. Twist both knobs back: the pins snap up into the shoe. Push the DS-014 plug into the payload's socket.

Hold point: no red band shows on either pin, and a firm tug does not move the shoe.

Step 11: into the frame

Step 11: into the frame, and the frame's power harness

Bolt the core under the deck; lid off, the frame's four leads in through the grommets, soldered to the pads, tied to the bar



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

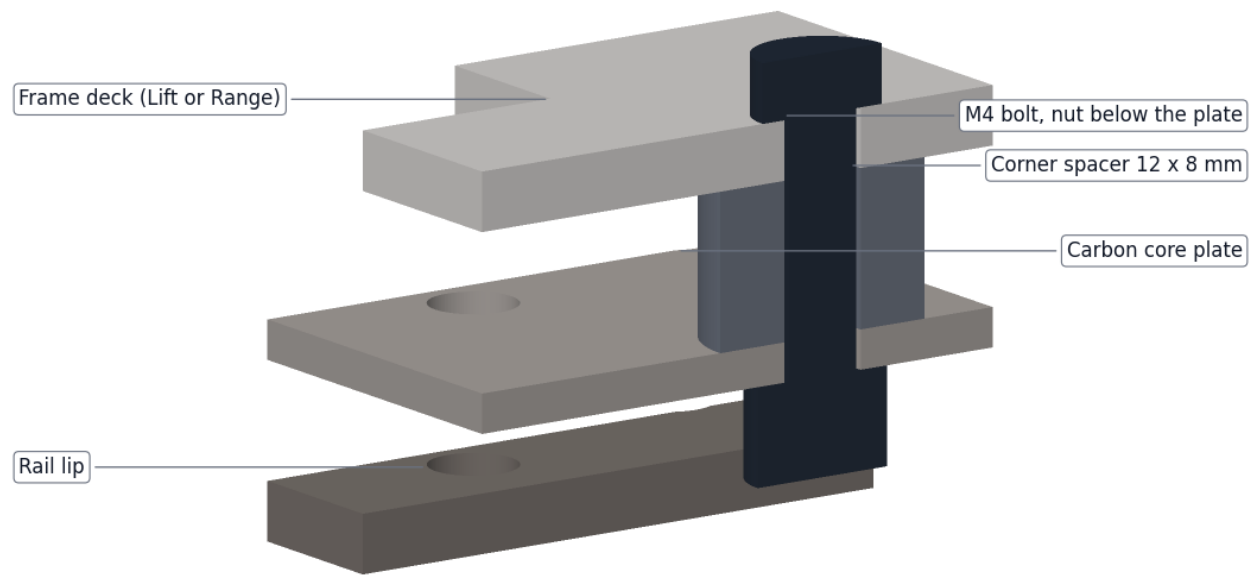
github.com/BoujeeEnjinia1701/kitewright-core

Put a corner spacer on each frame hole of the plate, lift the core so the lid passes up through the frame deck's opening, and fit four M4 bolts down through deck, spacers and plate with nylon-insert nuts on washers underneath.

Then fit the frame's power harness, which comes with Kitewright Lift or Range: four 8 AWG leads with AS150 plugs already on their far ends, inputs and outputs of opposite gender, labelled "PACK 1", "PACK 2", "FRAME A" and "FRAME B". Take the six nuts off and lift the lid off from above. Thread the leads in through the rear grommets, two per grommet, and solder each to its pad on the power board (pack inputs and frame outputs as the board marks them). Refit the lid. Tie the leads to the strain-relief bar with four cable ties, leaving a little slack between the bar and the grommets. Plug the frame's speed controller leads into the two frame outputs. To move the core to the other frame (R8 as restated on 2026-10-04, KWC-DDR-004), disconnect every pack, take the lid off, desolder the four leads, solder on the other frame's harness the same way and repeat safety stop 2; it takes about 30 min.

Joint 7: core to frame deck at a corner, cut through the bolt

The core hangs under the deck on four M4 bolts; the lid rises through the deck opening



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-core

Figure 28. Core to frame deck at a corner: the deck belongs to Kitewright Lift or Range.

■ 5. First checks

Table 3. First checks; a TRL 4 test report records the results.

Check	Requirement	How	Pass when
Flush underside	R2	Straight edge across the plate's underside after steps 3, 4, 7 and 9	No countersunk head stands proud
Rail fit	R2, R3	Slide the shoe the full rail length; feeler gauge above it	No binding; 0.2 to 0.4 mm gap; both pins drop in at the stops
Pin retention	R3	Pull each pin to its rest and back; tug the shoe with one pin withdrawn	Each pin alone stops the shoe; red band shows only when withdrawn
Proof load	R3	Hang 15 kg from the shoe for 1 min on the bench (core on a fixed stand, nobody beneath)	No permanent set, no slip
Payload swap	R2	Timed swap with winter gloves, three people, three times each	60 s or less, no tools
First power	R4, R5	Bench supply at 24 V, current limit 1 A, no packs	Controller boots; both 5.3 V rails in range; no part warm to the touch
Payload power	R4	Payload switch on into a 100 W load at 18 V and 50 V	Load holds 100 W; switch and pigtail stay below 60 °C
Cold boot	R5	Soak the core at -20 °C for 2 h, then power from the bench supply	Boots and logs normally
Interface	R1, R8	DS-014 continuity and a MAVLink payload heartbeat with a reference payload	Every pin in the table connected; payload recognised
Firmware	R10	Load the released PX4 build and the parameter set	No source changes; parameters load
Mass	R9	Weigh the core without the shoe, packs and frame harness	1.0 kg or less (0.99 kg estimated)

■ 6. Safety stops

Work stops at each point below until what it says is true.

1. **Before first power.** Every solder joint inspected; no pack anywhere near the bench; the power board's inputs checked for shorts with a meter; the bench supply set to 24 V and 1 A.
2. **Before connecting a pack.** The bench checks of section 5 passed; the frame harness's inputs and outputs keyed by opposite-gender AS150 plugs and labelled; polarity checked at the plug with a meter; the pack warm enough to discharge (ColdCell reports at least 10 °C); a lithium fire plan, sand bucket and extinguisher at hand; the core on a non-flammable surface; nobody alone.
3. **Before loading the mount.** Both pins show no red band and the shoe passes the tug test; the core is held on a fixed stand, not in a frame; nobody under the load.
4. **Before fitting propellers or arming any frame.** That frame's own build plan safety stops apply; the core's safety switch works; failsafes for the altitude band are loaded.
5. **Before any flight with a payload.** Pin check and tug test done by a second person; the flight follows local rules and the site partner's permissions.

■ 7. Tools, skills and workspace

- Hacksaw or bandsaw, files, deburring tool, square and scribe; pillar drill with 3.2, 3.4, 4.5, 5.5, 6 and 9.0 mm drills; 90° countersink; M10 x 1 tap and wrench.
- A small milling machine for the windows and pockets in the rail bars, or a local machine shop; anodising by a local shop.
- Thread-locker, a straight edge, and a 2-part epoxy for sealing the carbon plate's edges.
- FDM printer that can print ASA (enclosure, heated bed); heat-set insert tip.
- Temperature-controlled soldering station of 100 W or more for 8 AWG; heat-shrink and heat gun; crimp tools for JST-GH.
- Multimeter, current-limited bench supply (at least 60 V, 5 A), feeler gauges, kitchen scale.
- Skills: careful marking and drilling to 0.2 mm, tapping, soldering heavy leads, and following the PX4 setup guide.
- A bench with good light and ventilation, and a non-flammable area with a lithium fire plan for any work with packs.

■ 8. Where the numbers come from

- `cad/src/model.py` : every size and position, and the 720 constructability checks.
- `cad/drawings/KWC-DWG-001` : general arrangement (Rev P2); `KWC-DWG-101` to `109` : the making sketches.
- `docs/04-calcs/01-sizing.md` and `sizing.py` (`KWC-CAL-001`): loads, swap time, power, heat, altitude, mass and cost.
- `bom/bom.csv` : parts, specifications and prices.
- `docs/decisions/0001-trl2-review-decisions.md`, `0002-design-for-construction.md` and `0003-r9-core-mass.md`.